



UMESH

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EDUCATION

B.TECH(MECHANICAL ENGINEERING)	2023-2027	NSUT, New Delhi	6.98
CBSE(Class XII)	2022	Sarvodaya Boys S S S Matiala, New Delhi	84%
CBSE(Class X)	2020	Sarvodaya Boys S S S Matiala, New Delhi	82%

Summary

Software-focused Engineer with a solid foundation in the full software development lifecycle, specializing in backend engineering and full-stack web development. Proficient in **C++, Python, JavaScript**, and building robust RESTful APIs using the **MERN stack, SQL, and MongoDB**. Highly skilled in software design principles, core DSA, and object-oriented programming. Eager to leverage strong analytical and problem-solving skills to build scalable software applications as an SDE.

PROJECTS

Video Hosting Platform Backend | *Node.js, Express.js, MongoDB, Mongoose, JWT, Cloudinary*

- Developed a RESTful backend for a YouTube-like video hosting platform with user, video, comment, like, playlist, subscription, and watch-history functionality.
- Implemented JWT-based authentication with access/refresh tokens, secure password hashing, protected middleware, cookie-based sessions, and Cloudinary/Multer-based media uploads.
- Designed MongoDB schemas and aggregation pipelines, built modular routes/controllers with custom error handling, and tested APIs using Postman.

AI Desktop Assistant | *Python, Google Gemini API, ONNX TTS*

- Built a modular Python AI desktop assistant integrating Google Gemini API for conversational responses and natural-language command handling.
- Implemented Windows application automation and command routing to execute local system actions through user commands.
- Integrated ONNX-based local text-to-speech and structured the application into modular AI, automation, and voice-processing components.

ACADEMICS ACHIEVEMENTES

- Secured a position among the **Top 5** scorers in the Class X CBSE Board Examination at the school level.

TECHNICAL SKILLS

- Languages:** C++, Python, JavaScript, SQL
- Frontend:** React.js, Tailwind CSS
- Backend:** Node.js, Express.js, REST APIs
- Databases:** MongoDB, MySQL
- Tools & Technologies:** VS code Git, GitHub, Postman, JWT, Mongoose, Cloudinary, Vite
- Core Concepts:** Data structure and algorithm(DSA), Object oriented programming(OOP), Database management systems(DBMS)

POSITION OF RESPONSIBILITY

- ASME Student Chapter, NSUT — Recruitment Team Member**

Supported the student recruitment process through candidate interviews, applicant communication, and systematic organization, analysis, and shortlisting of candidate data using Excel.